

SPARSH SINGH

AI Systems Engineer • Full-Stack & Autonomous Agents Architect

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PROFESSIONAL SUMMARY

Systems and AI Engineer with deep expertise in multi-modal forensic AI, autonomous agent orchestration, edge computing, and distributed backend architectures. Proven record of engineering high-impact solutions for national-level challenges—including air-gapped border biometrics (Smart India Hackathon 2026) and deterministic FinTech operations (Paytm AI Hackathon 2026). Dedicated to deterministic guardrails, zero-hallucination policies, and 100% automated test coverage across production systems.

TECHNICAL SKILLS

Languages	Python (3.10+), TypeScript, JavaScript (ES6+), C++, SQL, HTML5/CSS3
AI / ML & Vision	PyTorch, ONNX Runtime, Vision Transformers (ViT), Wav2Vec 2.0, OpenCV, RapidOCR, AdaFace, MiniFASNet, Sarvam AI (sarvam-105b), XGBoost
Backend & DBs	FastAPI, Node.js, Express.js, Next.js (App Router), Redis (Pub/Sub), PostgreSQL (Prisma, Supabase), SQLCIPHER, SQLite (WAL Mode)
Cloud & DevOps	AWS (S3, SQS, DynamoDB), Docker, Render, Vercel, Meta WhatsApp Cloud API (v21.0), Cloudflare Tunnels, GitHub Actions (CI/CD)
Frontend & Browser	React, Next.js 14, Tailwind CSS, Monaco Editor, Chrome Extensions (Manifest V3), HTML5 Canvas, IndexedDB / Dexie.js, CRDTs
Architectures	Air-Gapped Zero-Trust Systems, Deterministic Policy Guardrails, Cryptographic Token Verification, TDD (Pytest, Jest)

FEATURED PROJECTS

NETRA — Enterprise Multi-Modal Forensic AI & Threat Intelligence Platform

FastAPI • Next.js 14 • AWS (S3, SQS, DynamoDB) • PyTorch • ViT • Wav2Vec 2.0 • Meta WhatsApp API

- Architected an institutional-grade cyber-defense platform combating generative AI deepfakes, digital arrest scams, and voice cloning, aligned with **Reserve Bank Innovation Hub (RBIH)** & **I4C** standards.
- Designed a **multi-modal late decision fusion pipeline** coupling spatial CNN boundary seam detectors with Vision Transformer (ViT) latent diffusion analyzers, Wav2Vec 2.0 acoustic spectrogram processors, and RapidOCR token extraction.
- Integrated **Meta WhatsApp Cloud API (v21.0)** for zero-barrier citizen triage reaching 535M+ users, backed by asynchronous AWS S3 + SQS worker nodes and DynamoDB job states.
- Automated compilation of statutory court-admissible FIR evidence citing **BNS 2023 Sec 318(4)** and **IT Act Sec 66D**; maintained **52/52 passing pytest suite (100% pass rate)** under adversarial SQL injection probes and concurrency stress tests.

Resolve OS — Autonomous AI Operations Teammate for Paytm Merchant Support

FastAPI • Sarvam AI (sarvam-105b) • Next.js 14 • Supabase PostgreSQL • WhatsApp Cloud API

- Built for the **Paytm ♥ AI Hackathon 2026**; automates tier-1 merchant dispute resolutions, payment reconciliations, and terminal operations in Hinglish.
- Devised a **dual-layer architecture** pairing Sarvam AI Hinglish NLU with a non-LLM mathematical deterministic policy engine to ensure zero AI hallucinations in financial execution.
- Enforced single-use **SHA-256 cryptographic action tokens** with pre-condition validation before executing ledger settlements, achieving **40/40 correct decisions and 0 unsafe financial actions**.
- Shipped a real-time reactive merchant operations center with Next.js 14, Tailwind CSS, and Supabase audit trails.

SSB Smart Document Screening — Air-Gapped Edge AI Identity Forensic Workstation

Python 3.11 • ONNX Runtime • AdaFace • MiniFASNet • SQLCipher • ICAO 9303 • UIDAI RSA-2048 PKI

- Engineered for **Sashastra Seema Bal (Smart India Hackathon 2026)** to secure Indo-Nepal and Indo-Bhutan border checkpoints in strictly air-gapped, zero-trust environments.
- Implemented an **ICAO Doc 9303 MRZ optical decoupler** (7-3-1 check digit algorithms) and an offline **UIDAI RSA-2048 PKI cryptographic signature verifier** for Aadhaar QR payloads.
- Integrated **AdaFace 1:1 facial matching** and **MiniFASNet multi-spectral liveness anti-spoofing** optimized via ONNX Runtime for edge CPU execution (<300ms inference).
- Utilized AES-256 encrypted **SQLCipher** for tamper-proof offline storage; verified with **54/54 passing tests**.

CodeRev — Real-Time Competitive Programming & Algorithmic Arena

Next.js 14 • Monaco Editor • Docker • Redis Pub/Sub • PostgreSQL 16 • Prisma ORM

- Developed an interactive competitive coding platform featuring 600+ algorithmic problems, AST-based memory visualizers, and multiplayer 1v1 battle rooms.
- Built an **ephemeral sandboxed multi-language remote execution engine** using isolated Docker containers with memory and CPU cgroups to prevent jailbreaks.
- Engineered dynamic **Glicko-2 / Elo matchmaking** and real-time multiplayer code synchronization via Redis Pub/Sub; verified by **48/48 passing tests**.

NullNote — Timestamped Video Knowledge Capture Chrome MV3 Extension

Chrome Manifest V3 • TypeScript • Next.js 14 • Shadow DOM • IndexedDB (Dexie) • CRDTs

- Engineered a Manifest V3 Chrome Extension and workspace for sub-second video telemetry, slide frame capture, and bi-directional note synchronization on YouTube.
- Pioneered a **two-channel delivery protocol** (`chrome.tabs.sendMessage` + `chrome.runtime.sendMessage`) to guarantee reliable state delivery across isolated in-tab iframe boundaries.
- Implemented offline-first storage using **IndexedDB (Dexie.js)** with CRDT conflict resolution and automated export to Notion, Obsidian, and Markdown (**42/42 passing tests**).

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

2023 – 2027
(Expected)

Key Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning.

HONORS & ACHIEVEMENTS

- **Paytm ♥ AI Hackathon 2026:** Architected Resolve OS with zero unsafe financial actions and 100% policy test coverage.
- **Smart India Hackathon (SIH 2026) Project Lead:** Engineered SSB air-gapped border document and identity screening workstation.
- **Open-Source & Agentic Workflows:** Authored open-source skills for Google DeepMind Antigravity and maintain 25+ public repositories on GitHub.